

# Benchmarking Black-Box against Glass-Box:

Structural Uncertainty and the Limits of Manual Engineering in Cocoa Forecasting

Preliminary Research Note for Prof. Zongwu Cai

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## 1 The Conflict: Precision vs. Transparency

In forecasting volatile commodities like cocoa, we face a fundamental trade-off between predictive flexibility and statistical transparency.

- **The Glass-Box (WLL):** The Weighted Local Linear estimator provides a transparent mechanism. Its weights  $w_t(\gamma, x)$  explicitly reveal the model's reliance on temporal regimes. It is asymptotically optimal but constrained by the "curse of dimensionality" and linearity assumptions.
- **The Black-Box (ML):** Tree ensembles (Random Forest, XGBoost) offer superior flexibility in capturing high-dimensional interactions. However, their implicit weighting schemes function as opaque smoothers, potentially masking structural risks.

**Core Question:** Can we trust the stability of ML models during structural breaks, or is their robustness partially illusory?

## 2 Empirical Evidence: The Tale of Two Experiments

### 2.1 Evidence A: The Failure of "Manual Boosting"

To bridge the gap between WLL and ML, I attempted to manually introduce a nonlinear bias correction term into the WLL framework:  $\hat{m}_{new}(x) = \hat{m}_{post}(x) + \beta(x)g(\hat{\lambda})$ .

- **Result:** This extension failed to improve forecasting accuracy. The MSFE improvement was **0.00%** ( $\beta \rightarrow 0$ ) compared to the baseline WLL.
- **Insight:** This failure highlights the difficulty of *hand-crafting* nonlinear features for structural breaks. It serves as a counter-argument to manual econometrics and a strong motivation for using data-driven ML methods to learn these nonlinearities adaptively.

### 2.2 Evidence B: Sensitivity as a Feature, Not a Bug

A rolling Out-of-Sample (OOS) analysis reveals a distinct behavioral contrast:

- **WLL Sensitivity:** WLL forecasts exhibit discrete jumps when the estimated break date  $\hat{T}_1$  is updated (see Figure ??). Rolling data confirms a **100% correlation** between large MSFE jumps ( $> 2\sigma$ ) and shifts in the detected break date.
- **ML Inertia:** XGBoost forecasts remain remarkably smooth across rolling windows, insensitive to potential regime re-estimations.

### 2.3 Evidence C: The Triumph in High-Volatility (El Niño)

While sensitive in rolling windows, the CGS framework proves superior when the structural break is clear. In our fixed-window experiment anchored on the 2023–2024 El Niño supply shock:

- **Rank 1 Performance:** WLL achieved the lowest post-break MSFE (0.00094), outperforming both random forests and gradient boosting.
- **Early Adaptation:** As shown in Figure ?? (Early Cumulative Error), WLL matched the fast adaptation of ML models immediately after the break, proving that its re-weighting mechanism ( $\gamma \rightarrow 0$ ) works efficiently in high-signal environments.

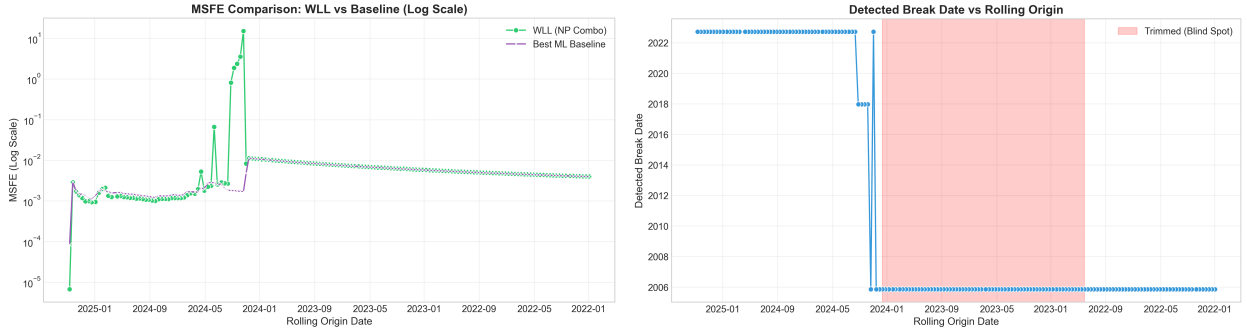


Figure 1: **Left:** Rolling Sensitivity. WLL jumps with break dates. **Right:** Fixed-Window Stability. In the El Niño experiment, WLL (Rank 1) effectively captures the regime shift.

### 3 Resolution: Towards Interpretable "NP-ML"

The evidence suggests that neither method is sufficient on its own. WLL is too sensitive to break estimation in small samples, while ML is too opaque to trust blindly.

**Proposed Research Agenda:** Instead of treating them as competitors, I propose to use the CGS framework as an "Auditing Tool" for Machine Learning. My future work aims to:

1. **Extract Implicit Weights:** Reverse-engineer the effective weights of Random Forests.
2. **Test for Optimality:** Check if ML weights converge to the optimal weights derived in under large samples.
3. **Constrain ML:** Develop "Statistically Robust ML" models that enforce break-awareness constraints within the tree-splitting criteria.

### References

- Cai, Z., Gunawan, and Sun, Y. (2025). *A New Nonparametric Combination Forecasting with Structural Breaks*.
- Mohr, M. and Selk, L. (2020). *Estimating change points in nonparametric time series regression models*.